

INDUSTRIAL INTERNSHIP REPORT**Project name: Smart File Organizer Using Python****Prepared By: Sree Varsha M**

1. Executive Summary

This project, Smart File Organizer Using Python, was developed during the Python Internship conducted by upskillCampus and UCT. The objective of the project is to automate file organization by categorizing files based on their extensions and moving them into appropriate folders. The project demonstrates the practical application of Python programming concepts such as file handling, conditional statements, loops, functions, and automation.

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3. 1. Preface

This internship provided practical exposure to Python programming and automation concepts. During the internship, I learned Python fundamentals, NumPy, Pandas, and file handling techniques. The project helped me understand how Python can be used to automate repetitive tasks and improve productivity.

4. 2. Introduction

The Smart File Organizer is a Python-based application that automatically organizes files into separate folders according to their file types. It reduces manual effort and helps users maintain a structured directory.

5. 3. Problem Statement

Managing large numbers of files manually is time-consuming and inefficient. Users often face difficulty locating files due to poor organization. The objective of this project is to automate the file organization process using Python.

6. 4. Existing and Proposed Solution

7. Existing Solution

Users manually create folders and move files individually, which is time-consuming and error-prone.

8. Proposed Solution

The proposed system automatically scans a selected directory, identifies file types, creates required folders, and moves files into their respective categories.

9. Code Submission (GitHub Link)

https://github.com/25it104-sreevarsha/upskillcampus_Python/blob/main/SmartFileOrganizer.py

10. Report Submission (GitHub Link)

https://github.com/25it104-sreevarsha/upskillcampus_Python/blob/main/SmartFileOrganizer_SreeVarsha_USC_UCT.pdf

11. 5. Proposed Design

12. High-Level Flow

13. User selects folder path.

14. Program scans files.

15. File extensions are identified.
16. Appropriate folders are created.
17. Files are moved automatically.

18. Categories

- Images
 - Documents
 - Videos
 - Audio
 - Archives
 - Others
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19. 6. Performance Test

20. Test Plan

Tested the application with multiple file types.

21. Test Procedure

- Created a sample folder with mixed files.
- Executed the Python program.
- Verified file categorization.

22. Performance Outcome

- Correct file categorization achieved.
 - Reduced manual effort.
 - Fast execution on standard systems.
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23. 7. My Learnings

Through this internship and project, I learned:

- Python Fundamentals
- Conditional Statements
- Functions
- File Handling
- Exception Handling
- NumPy Basics
- Pandas Basics
- Automation Concepts
- Debugging Techniques

The internship enhanced my programming and problem-solving skills.

24. 8. Future Scope

Future improvements can include:

- Graphical User Interface (GUI)
- Cloud Storage Integration
- Duplicate File Detection
- AI-Based File Classification
- Automatic Backup Features